# Mixed-Start Flow Matching for Action Generation

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**Abstract**

Flow-matching vision-language-action (VLA) models commonly generate action chunks starting from Gaussian noise. We decompose the source endpoint into two independently controllable factors—**conditional content** and **randomness**—and propose mixed-start flow matching. Learnable queries read a semantic start $z_0$ from a frozen VLM. During training, the source is $s = \beta h + (1-\beta)z_0 + \eta\varepsilon$ with $\beta \sim U(0, 0.3)$; deployment fixes $\beta = 0$. A mixed-source consistency loss avoids the supervision conflict caused by directly regressing the target action. Aggregated over three training seeds, the Gaussian/semantic paired ratios of repeat $\|\Delta\|$ are 2.80 on synthetic data and 2.04 on DROID, while the corresponding effective-action MSE ratios are 1.12 and 1.20; all cluster-bootstrap confidence intervals exclude 1. On synthetic held-out data from two training seeds, the ratios of optimization steps required to reach two early accuracy thresholds range from 2.5 to 5.0. Equal-noise-budget controls show that semantic content further reduces run-to-run divergence beyond noise reduction alone; the accuracy gain is significant on synthetic data, while the DROID effect has the same direction but its interval includes 1. Setting $\eta = 0$ at deployment eliminates source-sampling divergence without degrading accuracy. After training with a random Gaussian source, freezing one noise sample at deployment is also deterministic but significantly harms accuracy, yielding a per-trajectory paired MSE ratio of 1.31. Our conclusions are currently limited to one VLA family and open-loop replay. Step-count efficiency does not imply wall-clock acceleration, and fixed-noise retraining and real-robot closed-loop benefits remain to be validated.

**Keywords:** semantic start; mixed source; flow matching; action generation; run-to-run repeatability

## 1 Introduction

Flow matching [7] has become an important paradigm for continuous action generation. Conditioned on vision, language, and robot state, a VLA transports a source sample through a learned velocity field to produce a future action chunk [3, 4]. The standard approach uses Gaussian noise as the source, so resampling at each inference run introduces run-to-run variation. Simply freezing one noise sample removes this randomness but provides no task-dependent action prior.

History-based sources [1, 5, 10] exploit temporal continuity, whereas semantic sources [2] exploit current observations and instructions. We further separate two properties of a source endpoint: **what information it contains** and **whether it is random at deployment**. This decomposition leads to three contributions:

1. **A conditional semantic-start encoder.** $K = 64$ learnable queries pass through the frozen VLM. Their hidden states are mapped by an MLP and a zero-initialized output head to an action-chunk-shaped $z_0$. The semantic start enters only the flow source, not as an additional DiT token, preserving a clear separation from field-side conditioning.
2. **Mixed training with pure-semantic deployment.** Training mixes history and semantic starts only over $\beta \in [0, 0.3]$, while deployment fixes $\beta = 0$. A mixed-source consistency loss

supervises $z_0$ according to its role in the mixed source. The design does not cover the pure-history endpoint $\beta = 1$.

3. **Separate evaluation of source content and randomness.** Multiple training seeds, equal-noise-budget controls, fixed-noise deployment, and zero-noise deployment quantify the respective effects of semantic content, noise magnitude, and conditional determinism.

All primary comparisons share the same VLM and Action DiT backbones, data, optimization configuration, and evaluation protocol; the semantic-source arm additionally includes the conditional semantic-start encoder.

## 2 Related Work

**Flow matching and arbitrary sources.** Rectified Flow [8], Diffusion Policy [9], and GR00T [3] demonstrate the capability of iterative action generation. DDBM [11], I$^2$SB [12], and OT-CFM [13] show that the source distribution need not be Gaussian. We study a concrete source design for VLA action generation within this general framework.

**History and semantic sources.** A2A [10], UniWAM [5], and ΔFM [1] start from recent actions and exploit action continuity to reduce transport distance. UVM [2] produces a semantic start with a conditional encoder, and LeaP [14] further studies learnable source priors. We focus on the division between mixed training and pure-semantic deployment, and on how mixed-source consistency supervision avoids conflicting targets.

**Determinism and repeatability.** Deterministic prediction eliminates sampling divergence, but mean-squared regression tends toward the conditional mean when targets are multimodal. STAC [15] studies consistency monitoring for generative policies, and Flageat et al. [16] discuss return-level reproducibility. Our repeat $\|\Delta\|$ metric measures action divergence between repeated open-loop rollouts under the same initial conditions and is always reported together with accuracy.

## 3 Preliminaries

Given context $c$, a target action chunk $y \in \mathbb{R}^{H\times d_a}$, and a source $s$, the interpolation and velocity target at flow time $t$ are

$$x_t = (1-t)s + ty, \qquad u = y - s. \tag{1}$$

The model minimizes the mask-weighted loss

$$\mathcal{L}_{\mathrm{FM}} = \mathbb{E}\left[\|f_\theta(x_t, t, c) - (y-s)\|_M^2\right]. \tag{2}$$

Throughout, $t = 0$ **denotes the source and** $t = 1$ **the target**. Inference starts from $x_0 = s$ and uses $N$-step Euler integration:

$$x_{t+\delta} = x_t + \delta f_\theta(x_t, t, c), \qquad \delta = 1/N, \qquad \hat{y} = x_1. \tag{3}$$

We use $N = 4$ and action-chunk length $H = 40$. The context $c$ contains images, language instructions, and robot state; $h$ is the aligned previous action chunk; and $z_0 = E_\psi(c_{\mathrm{VL}})$ is generated solely from visual-language query features.

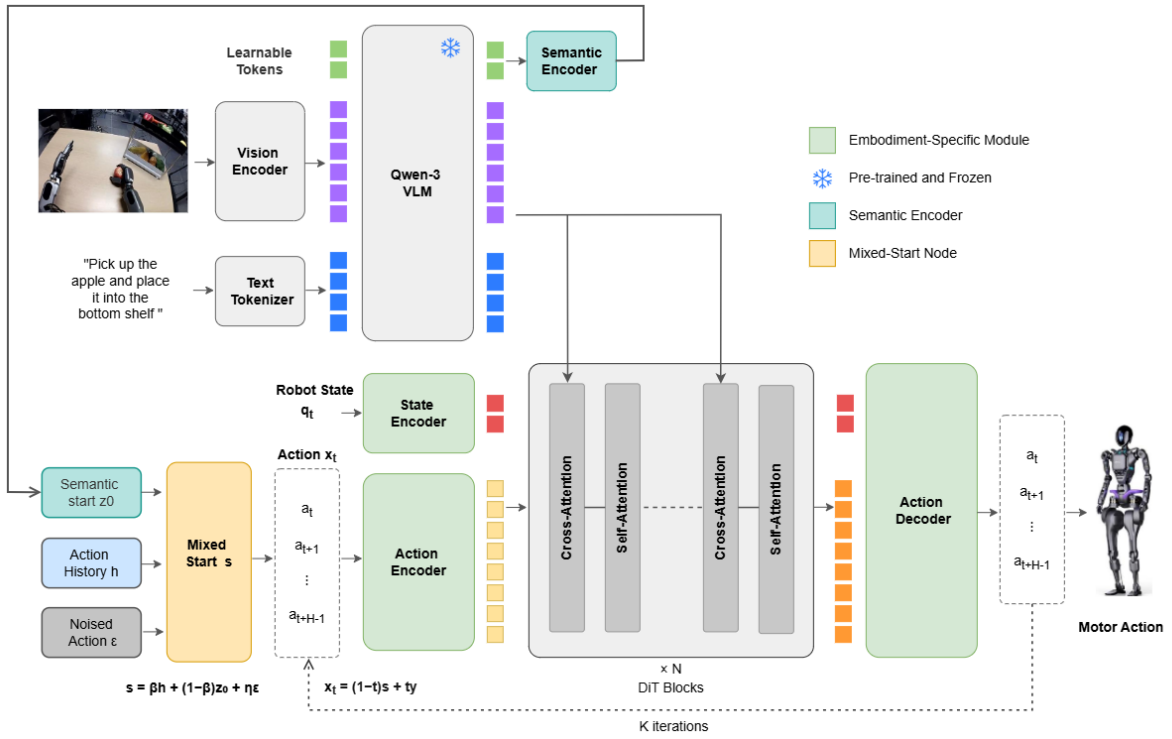



**Figure 1.** Mixed-start architecture. Images and text enter the frozen Qwen3-VL [17]. The final hidden states of learnable queries are mapped by the semantic encoder to $z_0$, which is combined with action history $h$ and noise $\varepsilon$ to form the source $s$. The interpolated action $x_t$ is encoded by the Action Encoder and processed by the Action DiT together with robot-state tokens and VLM features. The Action Decoder predicts the velocity field, and 4-step Euler integration produces the action chunk.

# 4 Method

## 4.1 Architecture and Conditional Semantic-Start Encoder

The semantic-start encoder has four components:

1. **Query injection.** $K = 64$ learnable queries $Q \in \mathbb{R}^{K\times 2048}$ are appended to the visual-language input embeddings and pass through the frozen backbone together with image and text tokens.

2. **Feature readout.** The final $K$ hidden states are processed by LayerNorm and a per-query linear projection.

3. **MLP mapping.** The query features are flattened and passed through a two-layer GELU-MLP and an embodiment-specific output head to produce $z_0 \in \mathbb{R}^{H\times D}$. The encoder has 178.7M stored parameters, but each sample activates only the output head for its embodiment, giving approximately 10.9M activated parameters.

4. **Zero initialization.** The output-head weights and biases are initialized to zero, so initially $z_0 = 0$ and no untrained random offset is introduced into the source.

The semantic start reads neither action history nor robot state. State-dependent local corrections are handled by the DiT state tokens, while the semantic encoder supplies an action sketch consistent with the task and observation.

### 4.2 Mixed Starts and Mixed-Source Consistency Loss

The history window uses strict-past alignment, shift/1: $h$ covers $[k-H, k-1]$ and does not overlap the target chunk $[k, k+H-1]$. The training source is

$$s = \beta h + (1-\beta) z_0 + \eta\, b_a \odot \varepsilon, \qquad \beta \sim U(0, 0.3), \tag{4}$$

where $\varepsilon \sim \mathcal{N}(0, I)$ and $b_a$ masks padding and near-discrete action dimensions. Training covers only weak history mixing and **does not include the pure-history source** $\beta = 1$.

At deployment, $\beta = 0$ removes only history mixing, leaving $s = z_0 + \eta\, b_a \odot \varepsilon$; setting $\eta = 0$ further yields the deterministic source $s = z_0$. **Source-sampling divergence** means that different inference runs start from different $s$, whereas **run-to-run divergence** refers to differences in the final predicted action sequences or rollouts. repeat $\|\Delta\|$ measures the latter and includes replanning and predicted-history accumulation. Neither concept denotes intersecting velocity fields or requires flow trajectories to cross.

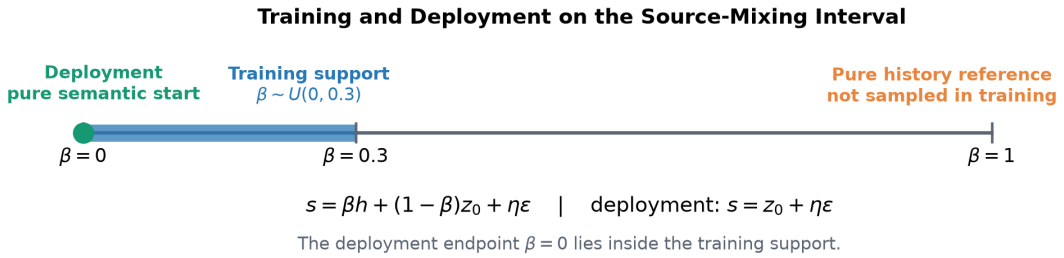



**Figure 2.** Training-deployment source division. The deployment endpoint $\beta = 0$ lies inside the training support; $\beta = 1$ denotes related history-source methods and is not sampled during training. The figure omits the action mask $b_a$ for compactness; the full definition appears in Eq. (4).

Directly minimizing $\|z_0 - y\|^2$ requires $z_0$ to approach the target action, whereas the flow loss through Eq. (4) requires it to compensate for the history term. To remove this conflict, we define the mixed-source consistency loss

$$\mathcal{L}_{\text{mix}} = \|\beta h + (1-\beta) z_0 - y\|_M^2. \tag{5}$$

Gradients propagate only through $z_0$; $h$ and $\beta$ are treated as constants. The total loss is

$$\mathcal{L} = \mathcal{L}_{\text{FM}} + \lambda(k)\mathcal{L}_{\text{mix}}, \tag{6}$$

where $\lambda(k)$ increases from 0 to 0.1 using a cosine schedule.

**Normalization and noise scale.** Target and history actions are first mapped to $[-1, 1]$ using per-dimension training-set quantiles $q_{01}$ and $q_{99}$. We then construct $z_0$ and $s$, with no further normalization or clipping after mixing. Thus, $\eta$ directly specifies the conditional noise standard deviation in normalized action space. The training value $\eta = 0.04$ references UniWAM's 0.02 noise scale and is twice as large; both use $[-1, 1]$ normalization, so no cross-range equivalence conversion is involved. We have not exhaustively swept the training-time $\eta$ and do not claim that 0.04 is optimal. Deployment with $\eta = 0$ is supported by the controlled results in Section 5.3.3.

## 5 Training and Evaluation

### 5.1 Data and Protocol

The primary comparisons initialize the Action DiT from scratch and load the remaining backbones from pretrained weights. The training set combines 300 DROID episodes [6] with 100

synthetic multi-segment episodes, totaling 400 episodes, 108,292 frames, and 17-dimensional actions. The synthetic data contain smooth within-segment motion, dwelling, instruction switches, and observation jumps to distinguish cases where history is useful from those where it becomes invalid. Data organization follows the action and embodiment conventions of Open X-Embodiment [18].

Evaluation uses:

- **Synthetic held-out data**: 10 independent 200-step episodes with segment-boundary annotations. A separate long-horizon check uses 10 independent 400-step episodes (8 segments of 50 steps), preserving the segment-length distribution.
- **DROID replay**: 30 episodes disjoint from the training set. The main tables use 10 of these episodes, while the zero-noise check uses all 30.

All evaluations use 4-step Euler integration and an execution window and replanning stride of 16 [19]. Repeatability evaluation starts two independent runs ($R = 2$) from the same initial observation for each trajectory; subsequent predicted histories may diverge. Paired confidence intervals use 20,000 cluster-bootstrap resamples [20]. The three-training-seed aggregate clusters by (training seed, trajectory), and other analyses cluster by trajectory. The main synthetic results aggregate four evaluation seeds unless a single evaluation implementation is explicitly stated.

## 5.2 Training Efficiency

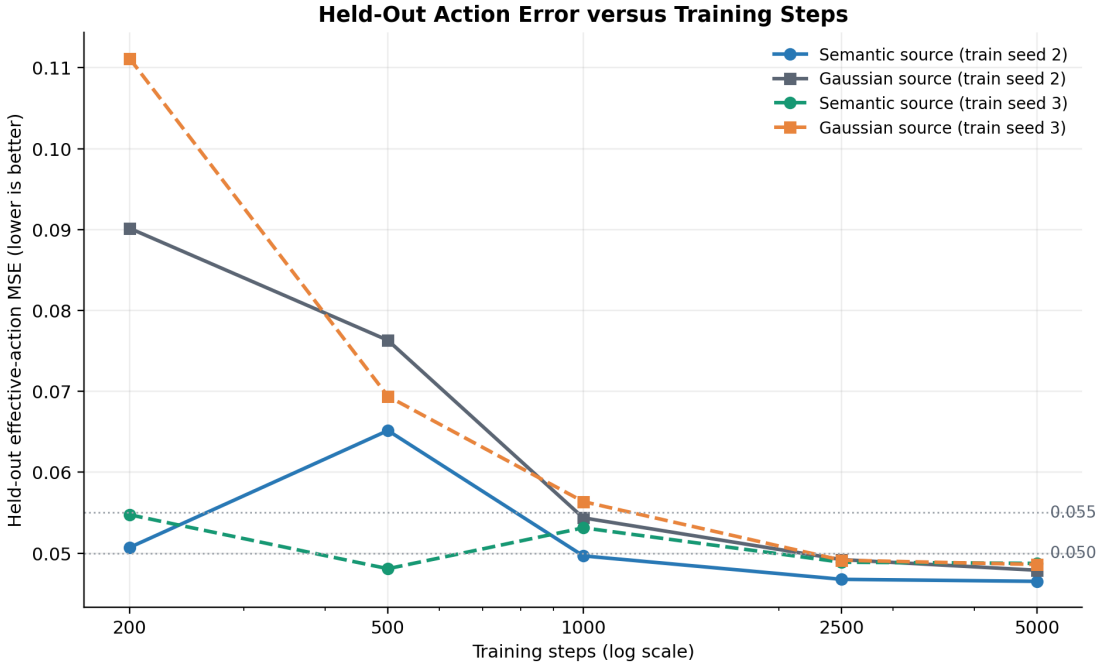



**Figure 3.** Open-loop held-out accuracy versus training steps. Independent comparisons with two training seeds show that the semantic source reaches usable accuracy at earlier checkpoints. Threshold-crossing steps are the first hits among discrete checkpoints, and the curves are not strictly monotonic; conclusions therefore use open-loop deployment metrics rather than training loss.

A second training seed reproduces the trend, with step ratios of at least 5 for both thresholds. On DROID, the Gaussian source requires 1000 steps to reach approximately the accuracy obtained by the semantic source at 200 steps. These results support improved **early training-step efficiency**, which is distinct from final accuracy. Because encoder overhead and wall-clock time are not included, we do not claim proportional compute savings.

**Table 1.** Training-step efficiency at early accuracy thresholds.

| Accuracy threshold | Semantic steps | Gaussian steps | Step ratio |
|---|---|---|---|
| mse_eff $\leq 0.055$ | 200 | 1000 | **5.0**× |
| mse_eff $\leq 0.050$ | 1000 | 2500 | **2.5**× |

Note: training seed 2, synthetic domain, evaluation seed 0, and $n = 10$; the step ratio is Gaussian/semantic.

## 5.3 Main Results

### 5.3.1 Metrics

- mse_eff: MSE between predicted and ground-truth actions after removing dimensions whose ground-truth standard deviation is below $10^{-3}$; lower is better.
- repeat $\|\Delta\|$: mean absolute action difference over all steps and action coordinates between two independent rollouts of the same trajectory; lower is better. It includes replanning and predicted-history accumulation, but differs from source-sampling sensitivity with fixed ground-truth history. With $R = 2$ and no action-scale normalization, it is neither a within-run variance estimate, a closed-loop success rate, nor a real-robot repeatability measure.
- **Paired ratio**: unless otherwise specified, ratios are Gaussian/semantic, so values above 1 favor the semantic source. At zero noise, repeat $\|\Delta\|$ is trivially zero and must be interpreted together with accuracy.

### 5.3.2 Accuracy and Run-to-Run Repeatability

**Table 2.** Accuracy and run-to-run repeatability over three training seeds.

| Metric | Synthetic | DROID replay |
|---|---|---|
| repeat $\|\Delta\|$ paired ratio | **2.80 [2.66, 2.95]** | **2.04 [1.95, 2.15]** |
| mse_eff paired ratio | **1.12 [1.06, 1.18]** | **1.20 [1.10, 1.30]** |

Note: brackets denote 95% cluster-bootstrap confidence intervals; ratios are Gaussian/semantic, and values above 1 favor the semantic source.

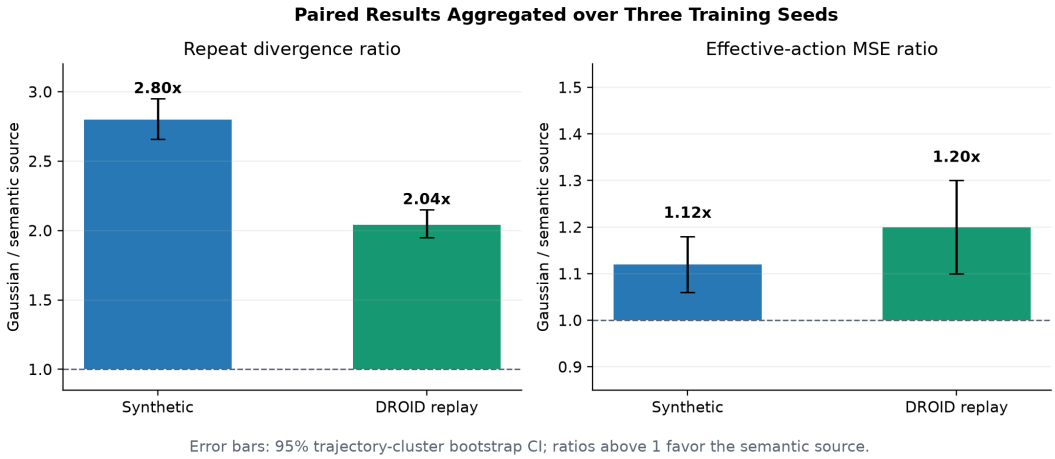



**Figure 4.** Paired results over three training seeds. Confidence intervals for both metrics in both domains lie on the favorable side of 1. These intervals describe paired uncertainty for the evaluated samples, not complete inference over an unlimited population of training seeds.

For training seed 2, semantic-source repeat $\|\Delta\|$ is 0.0278 on synthetic data and 0.0342 on DROID, compared with 0.0793 and 0.0688 for the Gaussian source. The corresponding mse_eff values are 0.0470 versus 0.0507 and 0.0192 versus 0.0227.

#### 5.3.3 Zero Noise, Observation Perturbations, and Long Horizons

**Table 3.** Zero-noise, observation-perturbation, and long-horizon results.

| Protocol | Semantic mse_eff, $\eta = 0$ | Control mse_eff | repeat $\|\Delta\|$ finding |
|---|---|---|---|
| Synthetic, 200 steps | **0.0460** | 0.0465 (semantic, $\eta = 0.04$) | Exactly 0 without observation perturbations when $\eta = 0$ |
| DROID-30 | **0.0186** | 0.0195 (semantic, $\eta = 0.04$) | Exactly 0 without observation perturbations when $\eta = 0$ |
| Synthetic, 400 steps | **0.0457** | 0.0564 (Gaussian) | Gaussian/semantic divergence ratio 2.92 |
| Synthetic, 200 steps + observation perturbations | **0.0459** | 0.0477 (Gaussian) | Gaussian/semantic divergence ratio 13.9 |

Note: training seed 2 and evaluation seed 0. Each row uses a different evaluation protocol and rows should not be compared numerically; lower mse_eff is better.

Without observation perturbations, the paired accuracy ratio of $\eta = 0$ relative to $\eta = 0.04$ is 0.99 [0.98, 1.00], supporting "no accuracy degradation" rather than "guaranteed improvement." Observation perturbations add pixel noise with standard deviation 8/255 and 2% relative state-vector noise at each replanning step. The two runs receive independent perturbations, so repeat $\|\Delta\|$ at $\eta = 0$ is no longer trivially zero and instead reflects sensitivity to observation noise. The zero-noise configuration retains accuracy and reduces divergence under perturbations; the current experiments do not support the hypothesis that residual source noise improves robustness. The independent 400-step check shows that the advantage persists over the longer horizon, but remains limited to one training seed and open-loop replay.

## 6 Ablations

### 6.1 Equal Noise Budget: Separating Noise Reduction from Semantic Content

We compare three sources: standard Gaussian $\varepsilon$, small Gaussian $0.04\varepsilon$, and semantic $z_0 + 0.04\varepsilon$. All use the same training seed and optimization configuration.

**Table 4.** Source-content ablation under equal injected-noise budgets.

| Domain | Source | repeat $\|\Delta\|$ ↓ | mse_eff ↓ |
|---|---|---|---|
| Synthetic | Standard Gaussian | 0.0792 | 0.0507 |
| Synthetic | Small Gaussian | 0.0509 | 0.0487 |
| Synthetic | Semantic | **0.0278** | **0.0469** |
| DROID | Standard Gaussian | 0.0688 | 0.0227 |
| DROID | Small Gaussian | 0.0469 | 0.0202 |
| DROID | Semantic | **0.0342** | **0.0192** |

Note: training seed 2; synthetic results aggregate four evaluation seeds, and DROID uses evaluation seed 0; $R = 2$, and lower is better for both metrics.

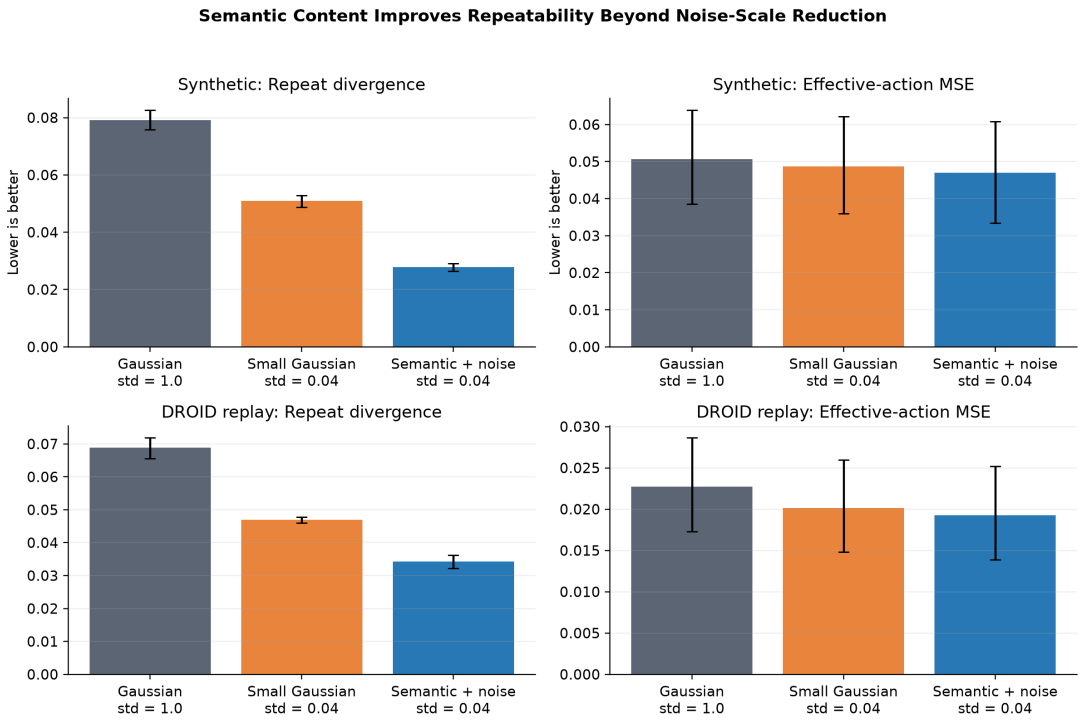



**Figure 5.** Equal-noise-budget control. A small Gaussian source already reduces divergence, while the semantic source further improves repeatability at the same injected-noise standard deviation.

As a descriptive decomposition of absolute means, noise reduction accounts for approximately 55% (synthetic) and 63% (DROID) of the total Gaussian-to-semantic reduction in repeat divergence. At the same $\eta = 0.04$, the semantic source further reduces repeat divergence by 45% and 27% relative to the small Gaussian source; both paired-ratio confidence intervals exclude 1. For accuracy, the semantic source significantly outperforms the small Gaussian source on synthetic data, while the DROID effect has the same direction but its interval includes 1. Thus, **noise magnitude is a major factor but does not fully explain the benefit of semantic content**. These percentages are not a causal variance decomposition, and the conclusion is limited to one training seed and the current protocol.

## 6.2 Is Fixed Noise Sufficient for Determinism?

After training with a random Gaussian source, freezing one Gaussian sample at deployment also reduces repeat $\|\Delta\|$ to zero, but the aggregate mean mse_eff increases from 0.0479 to 0.0564. The per-trajectory paired MSE ratio of frozen noise to normal resampling is 1.31, with 95% CI [1.07,1.62]. This ratio describes per-trajectory pairing and is not the percentage increase between the two aggregate means. By contrast, the semantic source with $\eta = 0$ retains both determinism and accuracy. This control tests only inference-time freezing after random training; it does not test fixed-noise retraining.

## 6.3 Mixed-Source Consistency Supervision

Compared with directly anchoring to $y$, mixed-source consistency supervision reduces the NMSE of $z_0$ tracking its consistency target from 5.46 to 2.04 at segment boundaries and from 4.13 to 0.97 within segments, reductions of 63% and 77%, respectively. This supports the motivation for Eq. (5): the supervision target should match the role of $z_0$ in the mixed source. This tracking comparison uses a pretrained-DiT configuration and is not directly comparable with

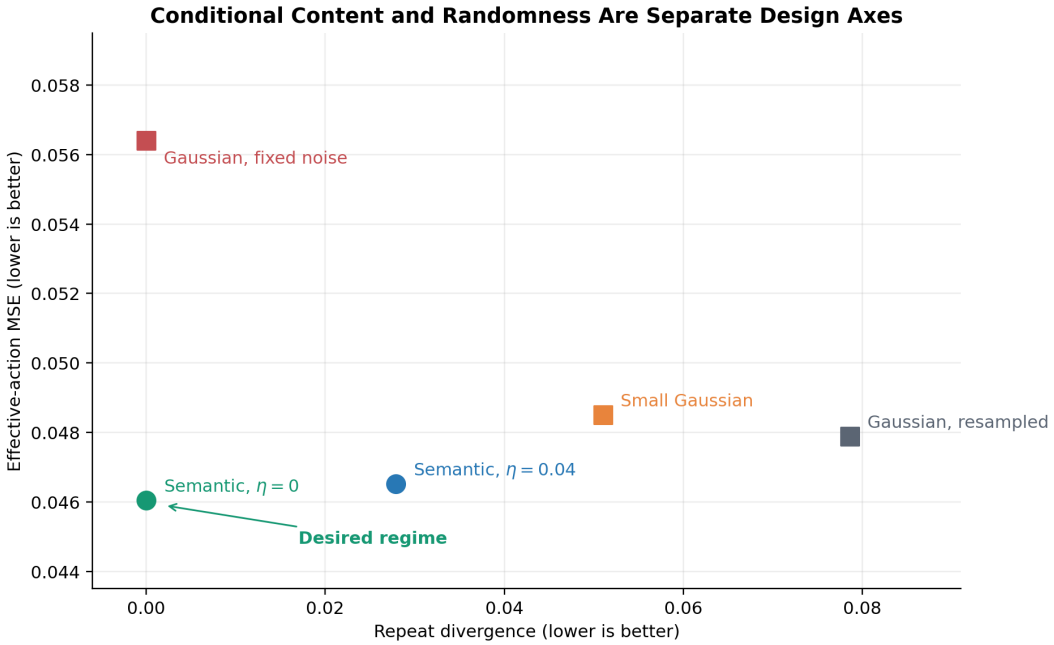



**Figure 6.** Joint comparison of source content and randomness. After random Gaussian training, inference-time noise freezing removes randomness only; the conditional semantic start provides both task-relevant content and determinism.

the from-scratch DiT results in Section 5.

In a from-scratch DiT comparison with training seed 3, annealed weighting ($\lambda$ increasing from 0 to 0.1 by a cosine schedule) outperforms constant weighting ($\lambda = 1$). The constant/annealed mse_eff ratios are 1.078 [1.046,1.112] on synthetic data and 1.426 [1.187,1.708] on DROID. Because this comparison changes both strength and schedule, it does not establish 0.1 as globally optimal or isolate either factor's independent contribution.

## 6.4 Encoder and History-Window Alignment

A $2 \times 2$ retraining comparison evaluates MLP versus Transformer encoders and strict-past shift/1 versus replanning-aligned rtc/16 history windows. All four arms use training seed 2 and otherwise identical configurations. Encoder capacities are unmatched, so the findings apply to these concrete configurations and do not constitute a multi-seed architectural law. The MLP and

**Table 5.** $2 \times 2$ retraining comparison of encoders and history-window alignment.

| Configuration | Synthetic mse_eff ↓ | Synthetic repeat↓ | DROID mse_eff ↓ | DROID repeat↓ |
|---|---|---|---|---|
| **MLP + shift/1** | **0.0469** | **0.0278** | **0.0192** | **0.0342** |
| Transformer + shift/1 | 0.0556 | 0.0483 | 0.0260 | 0.0566 |
| MLP + rtc/16 | 0.0689 | 0.0740 | 0.0292 | 0.0659 |
| Transformer + rtc/16 | 0.0889 | 0.1090 | 0.0418 | 0.1061 |

Note: training seed 2, 5000 steps, and the same evaluation protocol; encoder capacities are unmatched, and lower is better for all metrics.

strict-past window perform best in the current configuration. The rtc/16 history partially overlaps the target chunk and may create a training-time shortcut; this is a mechanism consistent with the results, not a completed causal proof. The ablation does not imply that Transformers are universally inferior to MLPs or that the conclusion transfers across model families.

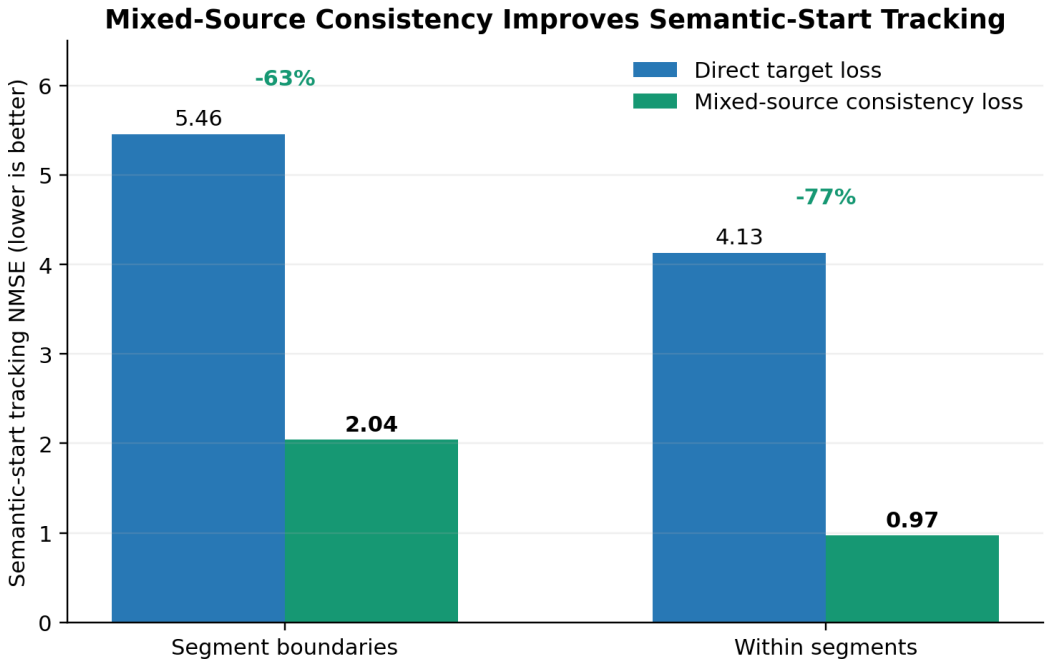



**Figure 7.** Effect of mixed-source consistency supervision on semantic-start tracking error. NMSE decreases both at segment boundaries and within segments; this result does not establish improved closed-loop success.

## 7 Conclusion

We presented an action flow-matching method composed of a conditional semantic start, weak history mixing during training, and mixed-source consistency supervision. The main evidence supports three findings:

1. Aggregated over three training seeds, the semantic source improves both action accuracy and run-to-run repeatability in two data domains.
2. On synthetic held-out data from two training seeds, the semantic start reduces the early training steps required to reach specified accuracy thresholds.
3. A zero-noise semantic source provides a deterministic, accurate deployment endpoint. In the current randomly trained Gaussian arm, inference-time noise freezing cannot replace conditional content; fixed-noise retraining remains untested.

The equal-noise-budget controls further bound the conclusion: repeatability gains arise from both noise reduction and semantic content and cannot be attributed entirely to the encoder.

## 8 Future Work

1. Validate the relationship between repeat $\|\Delta\|$, task success, and transition stability in real-robot closed-loop tasks.
2. Test architectural generalization on other VLA backbones and public closed-loop benchmarks such as LIBERO [21].
3. Study low-rank, temporally shared semantic decoding heads to reduce the parameter cost of embodiment-specific output heads.

4. Recalibrate the scope of zero-noise deployment under multimodal action distributions and out-of-distribution perturbations.
5. Compare deterministic direct regression, constant-source retraining, and fixed-noise retraining under the same protocol.

# A Theoretical Notes

## A.1 Conditional Mean and Deterministic Prediction Error

If $z_0$ is directly trained with an MSE target $y$, its conditional optimum is $\mathbb{E}[y \mid c]$, and the mean-squared error lower bound for deterministic conditional prediction is $\mathbb{E}\,\mathrm{Var}(y \mid c)$. This bound describes irreducible uncertainty under the conditioning information, not an upper bound on generative-model quality. Our $z_0$ is supervised by mixed-source consistency and approaches the conditional mean as an ideal limit only when the upper endpoint of the mixing interval tends to zero; the velocity field performs the correction from the conditional start to the target action.

## A.2 Ideal Equilibrium of Mixed-Source Consistency Supervision

Let $\beta \sim U(0, b)$, independently of $(h, y)$. Taking the conditional stationary point of Eq. (5) with respect to $z_0$ gives

$$z_0^*(c) = \frac{(1 - b/2)\mathbb{E}[y \mid c] - (b/2 - b^2/3)\mathbb{E}[h \mid c]}{1 - b + b^2/3}. \tag{7}$$

As $b \to 0$, $z_0^*(c) = \mathbb{E}[y \mid c]$. In general, the optimum under mixed-source consistency supervision reflects both the target action and the history term, matching its actual role in the mixed source.

## A.3 Noise Standard Deviation

For fixed $h, z_0, \beta$, let $m = \beta h + (1 - \beta)z_0$. Ignoring the per-dimension mask for this calculation,

$$s = m + \eta\varepsilon \quad \Longrightarrow \quad s \mid m \sim \mathcal{N}(m, \eta^2 I). \tag{8}$$

Thus, $s = 0.04\varepsilon$ has per-dimension standard deviation 0.04 and variance 0.0016. Relative to a standard Gaussian source, its standard deviation is 25 times smaller and its variance is 625 times smaller. No second normalization is applied after mixing, so this noise budget is consistent between training and inference.

## A.4 Benefits and Limits of Conditional Transport Energy

Compare the small Gaussian source $s_{G\text{-small}} = a\varepsilon$ with the semantic source $s_Z = z_0(c) + a\varepsilon$. Let $\mu_y(c) = \mathbb{E}[y \mid c]$, and let $D$ be the number of coordinates actually receiving noise. Then

$$\mathbb{E}[\|y - s_{G\text{-small}}\|^2 \mid c] = \mathrm{tr}\,\mathrm{Cov}(y \mid c) + \|\mu_y(c)\|^2 + a^2 D, \tag{9}$$

$$\mathbb{E}[\|y - s_Z\|^2 \mid c] = \mathrm{tr}\,\mathrm{Cov}(y \mid c) + \|\mu_y(c) - z_0(c)\|^2 + a^2 D. \tag{10}$$

The reduction in conditional transport energy due to the semantic source is therefore

$$\Delta_{\mathrm{tr}}(c) = \|\mu_y(c)\|^2 - \|\mu_y(c) - z_0(c)\|^2 = 2z_0(c)^\top \mu_y(c) - \|z_0(c)\|^2. \tag{11}$$

We have $\Delta_{\mathrm{tr}}(c) > 0$ if and only if $\|z_0(c) - \mu_y(c)\| < \|\mu_y(c)\|$. This explains why a good semantic start may reduce initial target energy and the learning burden under finite budgets. However, the two sources have identical conditional velocity covariance, and reduced transport energy does not guarantee lower final action risk, fewer integration steps, or lower run-to-run divergence. Those outcomes still depend on field approximation, optimization, numerical integration, and the deployment protocol.

# B Reproducibility

**Table 6.** Primary training configuration.

| Item | Configuration |
|---|---|
| Action DiT | 1091.7M parameters, initialized from scratch; output projections zero-initialized |
| Optimizer and precision | AdamW, bfloat16 |
| Learning rate | $10^{-4}$ |
| Batch | Pre-accumulation batch size 4, gradient accumulation 2, effective batch size 8 |
| Training steps | 5000; training-efficiency analysis additionally evaluates checkpoints at 200, 500, 1000, and 2500 steps |
| Trainable parameters | 1799.2M |
| Semantic encoder | 178.7M stored; approximately 10.9M activated per sample |

The main results use the 5000-step checkpoint. Training-efficiency analysis uses the first threshold hit among discrete checkpoints rather than selecting the best test checkpoint.

- Code: `https://github.com/GUOZHIWEN/GR00T-N1.7-MixedStart`
- Weights: `https://www.modelscope.cn/models/joyogzw/GR00T-N1.7-MixedStart`
- Data: GR00T-N1.7-MixedStart-Datasets (ModelScope)
- Main implementation: `gr00t/model/gr00t_n1d7/`
- Unified open-loop evaluator: `gr00t/eval/compare_flow_sources.py`

# C Action Trajectory Comparison

Figures 8–9 visualize actual action outputs from the semantic-start and Gaussian-source models. Both arms use from-scratch Action DiT initialization, the same training data, training seed 2, and a 5,000-step training budget. Evaluation uses open-loop replay with evaluation seed 0. Black denotes ground truth, blue the semantic-start predictions, and gold the Gaussian-source predictions; action dimensions are fixed at 0, 1, 3, and 4.

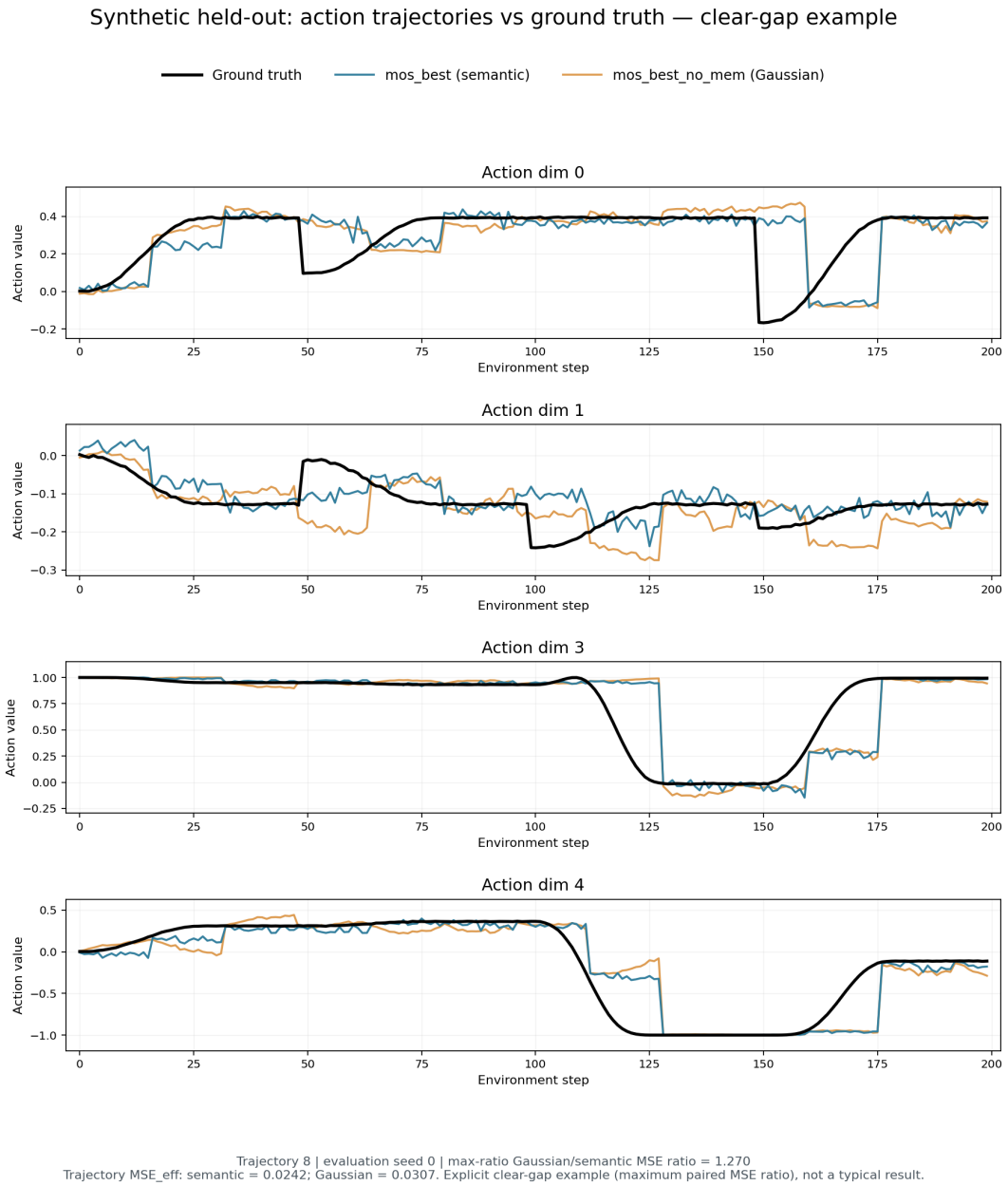



**Figure 8.** Synthetic held-out trajectory 8. Effective action MSE is 0.0242 for the semantic source and 0.0307 for the Gaussian source. This example has the largest Gaussian/semantic MSE ratio among 10 trajectories under the same protocol. It illustrates a clear-gap case, not the typical gain.

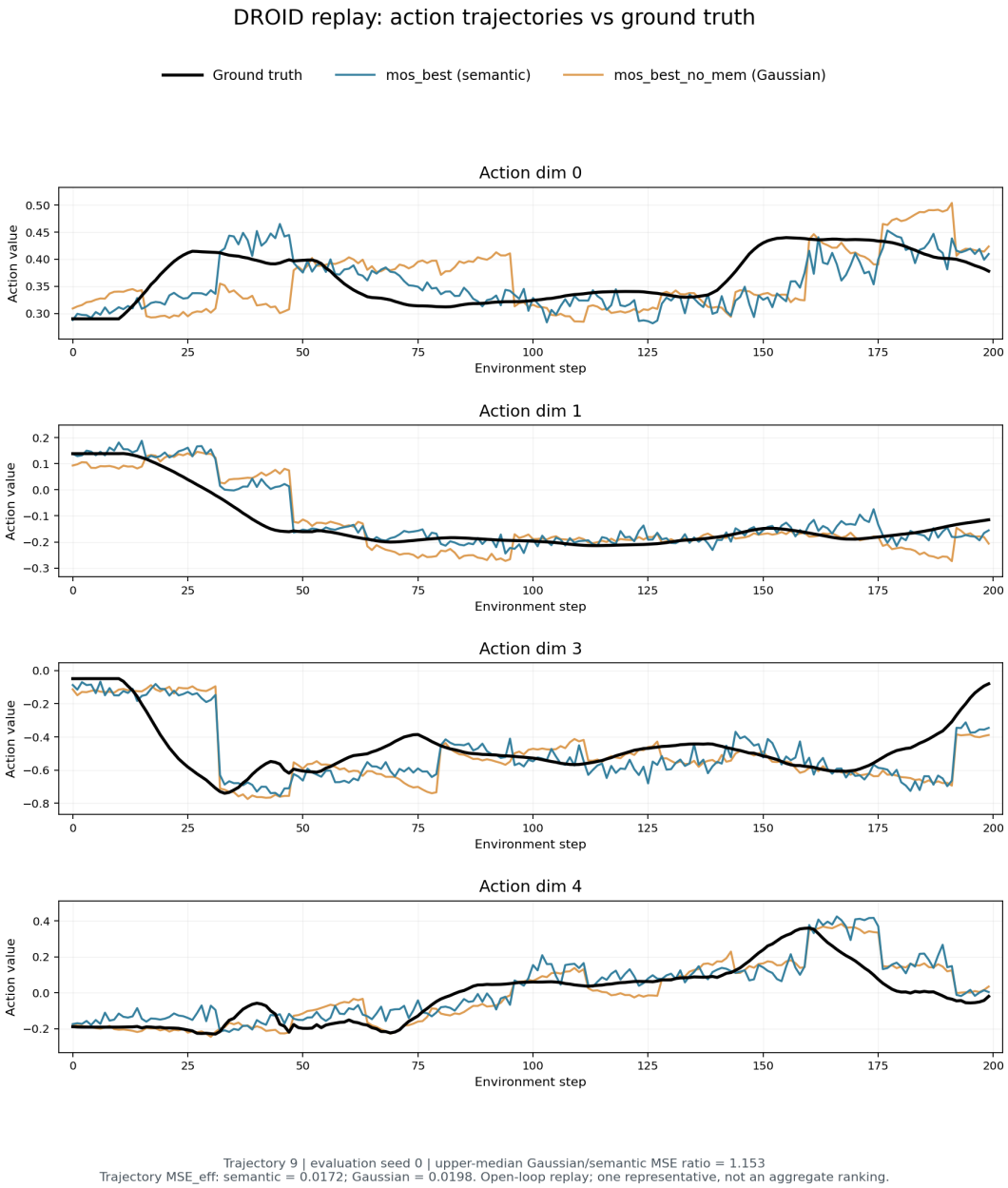



**Figure 9.** DROID replay representative trajectory 9. Selected by the upper-median Gaussian/semantic MSE ratio, this example has effective action MSE values of 0.0172 and 0.0198 for the semantic and Gaussian sources, respectively. Individual trajectories do not replace multi-seed paired statistics, and open-loop replay does not measure closed-loop task success.

## AI Assistance Disclosure

Generative AI tools were used for language polishing and formatting checks. The author is responsible for the method, experiments, data, and conclusions.

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